Faculty Initial: SBB

Dept. of CSE

School of Engineering

Date: Jan 27, 2026 12:00 PM

Examination	: Final Examination	**Semester**	: Fall 2025
Course Code	: MAT261	**Marks**	: 30
Course Title	: Linear Algebra and Matrices	**Section**	: 1, 2, 3, 4, 5
Slot	: Slot - 8	**Duration**	: 2 hours

General Instructions:

- All questions must be answered.
- All the devices (mobile, smartwatch, etc) should be turned off and kept in the student's bag during the exam period.

Sl. No.	Tasks	Marks	COs	BCL*
1.	**Explain** whether or not $W = \{(x, y, z) \in \mathbb{R}^3 : 3x - y - 4z = 0\}$ is a subspace of $\mathbb{R}^3$.	3	CO1	L-2 [Understand]
2.	**Determine** the coordinate vector of $v = 3x^3 - 4x^2 + 2x - 5$ relative to S, where $S = \{(x-2)^3, (x-3)^2, (x-1), 2)\}$	4	CO1	L-2 [Understand]
3.	**Show** whether the following Transformation is Linear or not: $T: \mathbb{R}^3 \to \mathbb{R}^2; T(x, y, z) = (x - y + 3z, 3x + 2y + 4z)$	3	CO1	L-3 [Apply]
4.	**Examine** whether the following transformation is invertible. If it is, find its inverse. Where $T = (x + 3y - z, 2x - 2y + z, 3x + 2y + 3z)$.	5	CO4	L-4 [Analyze]
5	**Express** the matrix representation of the linear operator $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (3y + z, x - 4y, 4x - 3y)$ with respect to the Basis: $B = \{(1,1,2), (1,1,3), (1,-1,-2)\}$	6	CO4	L-3 [Apply]
6.	Let, $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ (a) **Compute** the eigenvalues and corresponding eigenvectors of A (b) Construct the matrix P & **determine** $P^{-1}AP$ (c) Using the formula $A^n = PD^nP^{-1}$, **compute** A^{10}	4+3+2	CO3	L-3 [Apply]

5, 1

$\begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix}$ $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

2.4×10^6

7.3×10^6

*Bloom's Cognitive Level